

Active Directory PowerShell Cheat Sheet

Top cmdlets, syntax, and practical examples

Getting Started

Import the Active Directory module and get help.

Syntax:

```
Import-Module ActiveDirectory Get-Command -Module ActiveDirectory Get-Help Get-ADUser -Online
```

Get-ADUser

Retrieve users. Use -Filter to query and -Properties to include attributes.

Syntax:

```
Get-ADUser [-Identity] [-Filter ] [-Properties ] [-SearchBase ]
```

Examples:

- `Get-ADUser -Identity jdoe`
- `Get-ADUser -Filter * -Properties DisplayName,EmailAddress | Select SamAccountName,DisplayName,EmailAddress`
- `Get-ADUser -Filter {Enabled -eq $true} -Properties LastLogonDate`
- `Get-ADUser -Filter * -SearchBase 'OU=Sales,DC=contoso,DC=com'`

New-ADUser / Set-ADUser / Remove-ADUser

Create, modify, and delete user accounts.

Syntax:

```
New-ADUser -Name -SamAccountName [-UserPrincipalName ] [-Path ] Set-ADUser -Identity [-EmailAddress ] [-Enabled ] ... Remove-ADUser -Identity
```

Examples:

- `New-ADUser -Name 'Jane Doe' -SamAccountName jdoe -UserPrincipalName jdoe@contoso.com -Path 'OU=Sales,DC=contoso,DC=com'`
- `Set-ADUser -Identity jdoe -EmailAddress jane.doe@contoso.com -Enabled $true`
- `Remove-ADUser -Identity temp.user -Confirm:$false`

Set-ADAccountPassword / Unlock-ADAccount / Enable-ADAccount / Disable-ADAccount

Password and state management for accounts.

Syntax:

```
Set-ADAccountPassword -Identity -NewPassword (Read-Host -AsSecureString) -Reset
Unlock-ADAccount -Identity Enable-ADAccount -Identity Disable-ADAccount -Identity
```

Examples:

- Set-ADAccountPassword -Identity jdoe -NewPassword (ConvertTo-SecureString 'P@ssw0rd!' -AsPlainText -Force) -Reset
- Unlock-ADAccount -Identity jdoe
- Disable-ADAccount -Identity test.user

Get-ADGroup / New-ADGroup / Set-ADGroup / Remove-ADGroup

Query and manage groups.

Syntax:

```
Get-ADGroup [-Identity] [-Filter ] [-Properties ] New-ADGroup -Name -GroupCategory
-GroupScope -Path Set-ADGroup -Identity [-Description ] [-ManagedBy ] Remove-ADGroup
-Identity
```

Examples:

- Get-ADGroup -Filter {GroupCategory -eq 'Security'}
- New-ADGroup -Name 'Sales-Team' -GroupCategory Security -GroupScope Global -Path 'OU=Groups,DC=contoso,DC=com'
- Set-ADGroup -Identity 'Sales-Team' -Description 'Sales department members'
- Remove-ADGroup -Identity 'Obsolete-Group' -Confirm:\$false

Add-ADGroupMember / Remove-ADGroupMember / Get-ADGroupMember

Manage group memberships.

Syntax:

```
Add-ADGroupMember -Identity -Members Remove-ADGroupMember -Identity -Members
-Confirm:$false Get-ADGroupMember -Identity [-Recursive]
```

Examples:

- Add-ADGroupMember -Identity 'Sales-Team' -Members jdoe, msmith
- Get-ADGroupMember -Identity 'Sales-Team' -Recursive
- Remove-ADGroupMember -Identity 'Sales-Team' -Members temp.user -Confirm:\$false

Get-ADComputer / New-ADComputer / Set-ADComputer / Remove-ADComputer

Manage computer objects.

Syntax:

```
Get-ADComputer [-Identity] [-Filter ] [-Properties ] New-ADComputer -Name -SamAccountName
-Path Set-ADComputer -Identity [-Description ] [-Enabled ] Remove-ADComputer -Identity
```

Examples:

- `Get-ADComputer -Filter * -Properties IPv4Address | Select Name, IPv4Address`
- `New-ADComputer -Name 'PC001' -SamAccountName 'PC001$' -Path 'OU=Workstations,DC=contoso,DC=com'`
- `Set-ADComputer -Identity 'PC001' -Description 'Finance workstation'`

Get-ADOrganizationalUnit / New-ADOrganizationalUnit / Set-ADOrganizationalUnit / Remove-ADOrganizationalUnit

Query and manage OUs.

Syntax:

```
Get-ADOrganizationalUnit -Filter [-SearchBase ] New-ADOrganizationalUnit -Name -Path
[-ProtectedFromAccidentalDeletion $true] Set-ADOrganizationalUnit -Identity [-Description ]
[-ProtectedFromAccidentalDeletion <$true|$false>] Remove-ADOrganizationalUnit -Identity
```

Examples:

- `Get-ADOrganizationalUnit -Filter *`
- `New-ADOrganizationalUnit -Name 'HQ' -Path 'DC=contoso,DC=com' -ProtectedFromAccidentalDeletion $true`
- `Set-ADOrganizationalUnit -Identity 'OU=HQ,DC=contoso,DC=com' -Description 'Headquarters OU'`

Get-ADDomain / Get-ADForest / Get-ADDomainController

Discover domain, forest, and domain controller information.

Syntax:

```
Get-ADDomain [-Identity ] [-Server ] Get-ADForest [-Identity ] [-Server ]
Get-ADDomainController [-Filter ] [-Discover]
```

Examples:

- `Get-ADDomain`
- `Get-ADForest | Select Domains,GlobalCatalogs`
- `Get-ADDomainController -Discover -Service 'PrimaryDC'`

Move-ADObject / Rename-ADObject

Move or rename any AD object.

Syntax:

```
Move-ADObject -Identity -TargetPath Rename-ADObject -Identity -NewName
```

Examples:

- `Move-ADObject -Identity 'CN=PC001,OU=Workstations,DC=contoso,DC=com' -TargetPath 'OU=Finance,DC=contoso,DC=com'`
- `Rename-ADObject -Identity 'CN=TempUser,OU=HR,DC=contoso,DC=com' -NewName 'John.Temp'`

Add/Get/Remove-ADDomainControllerPasswordReplicationPolicy

Manage the allowed/denied lists for RODC password replication.

Syntax:

```
Add-ADDomainControllerPasswordReplicationPolicy -Identity -AllowedList
Get-ADDomainControllerPasswordReplicationPolicy -Identity -Policy
Remove-ADDomainControllerPasswordReplicationPolicy -Identity -DeniedList
```

Examples:

- `Get-ADDomainControllerPasswordReplicationPolicy -Identity RODC01 -Policy Allowed`
- `Add-ADDomainControllerPasswordReplicationPolicy -Identity RODC01 -AllowedList 'CONTOSO\BranchUsers'`

Fine-Grained Password Policies

Create and assign per-group/user password policies.

Syntax:

```
New-ADFineGrainedPasswordPolicy -Name -ComplexityEnabled $true -MinPasswordLength 14
-MaxPasswordAge 30.00:00:00 Add-ADFineGrainedPasswordPolicySubject -Identity -Subjects
Get/Set/Remove-ADFineGrainedPasswordPolicy
```

Examples:

- `New-ADFineGrainedPasswordPolicy -Name 'Admins-Strong' -MinPasswordLength 16 -MaxPasswordAge 15.00:00:00 -ComplexityEnabled $true`
- `Add-ADFineGrainedPasswordPolicySubject -Identity 'Admins-Strong' -Subjects 'CONTOSO\Domain Admins'`

Enable-ADOptionalFeature / Disable-ADOptionalFeature

Turn on/off optional directory features (e.g., Recycle Bin).

Syntax:

```
Enable-ADOptionalFeature -Identity 'Recycle Bin Feature' -Scope ForestOrConfigurationSet
-Target 'contoso.com' Disable-ADOptionalFeature -Identity -Scope -Target
```

Examples:

- `Enable-ADOptionalFeature -Identity 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -Target 'contoso.com'`

Tips & Best Practices

• Always run with least privilege; avoid Domain Admin for routine reads. • Use -Server to target a specific DC for consistency. • Use -SearchBase to scope queries (speeds up large forests). • Pipe to Select-Object for clean exports; use Export-Csv for reports.